

Practical Exercise 1: Data Types and Storage Architecture

Part 1 : Data Exploration

- In Kaggle website, select a dataset for each data type:
 - **Structured**
 - **Semi-structured**
 - **Unstructured**

Part 2 : Data Storage Design

- Outline how each dataset would be stored in a Data Warehouse vs. a Data Lakehouse.
- Identify any transformations needed for analysis in each architecture.

Part 3 : Comparison Analysis

- Create a table of pros/cons for storing and analyzing each data type in a Data Warehouse vs. Lakehouse.

Part 4 : Summary & Discussion

- Summarize findings on the flexibility and suitability of each architecture.

Practical Exercise 2: Data Modelling

Prerequisite

- Open the file : adventureworks Sales.xlsx

Part 1: Data Analysis

- **Objective:** Explore and understand the structure and content of the **AdventureWorks Sales.xlsx** file.
 - **Task:** Review each table to understand its purpose and the types of data it contains.
 - **Questions :**
 - 1) What is the role of each table in the sales process?
 - 2) Which fields seem to uniquely identify records, and which fields might link tables together?

Part 2: Key Identification

- In each table, identify:
 - **Primary Key:** The unique identifier for each record.
 - **Foreign Keys:** Keys that link to other tables, establishing relationships.

Part 3: Entity-Relationship Diagram (ERD)

- Using **draw.io**, create an ERD that includes:
 - Each table as an entity, listing columns.
 - Underline primary keys and use arrows to indicate foreign keys linking tables.

Part 4: Relationship Analysis

- Describe the relationships between tables as One-to-Many, Many-to-One or Many-to-Many.

Practical Exercise 3: Data Architecture based on business needs

Context:

Three companies A, B and C., each with specific business needs, are considering implementing a new data management architecture to enhance their analytics and their decision-making processes.

You will assess each company's requirements and recommend one of the following architectures: Data Warehouse, Data Lake, or Data Lakehouse.

About the Companies:

Company A – Healthcare Provider

- **Their needs :**
 - Store highly structured, sensitive medical data.
 - Ensure compliance with strict security and privacy regulations (such as GDPR for Europe).
 - Provide near real-time reporting to support clinical decision-making.
- **Their challenges :**
 - Requires high data security and strict governance.
 - The data volume is large, though primarily structured.

Company B - E-commerce Company

- **Their needs :**
 - Analyze large volumes of customer data from various sources (transactions, social media, customer reviews)
 - Real-time processing is needed for personalized product recommendations while customers browse.
 - Integration of semi-structured and unstructured data (e.g., customer reviews, social media interactions).
- **Their challenges :**
 - Data is diverse and voluminous, requiring flexible storage and strong real-time processing capabilities.
 - The company wants to incorporate machine learning in the future for customer recommendations.

Company C - Environmental Firm

- **Their needs :**
 - Store continuous environmental data from sensors and satellite imagery.
 - Manage massive amounts of unstructured data (images, video, sensor data).
 - Enable historical data analysis to monitor environmental trends over several years.
- **Their challenges :**
 - Large amounts of unstructured data need to be accessible for advanced analyses and modeling.
 - Retention of multiple historical versions of data is crucial for long-term environmental studies.

Instructions

Part 1: Analyze Each Company's Needs

1. For each company, identify and answer:
 - What types of data does the company manage?
 - What level of security and compliance is required?
 - Does the company need real-time processing, batch processing, or both?
 - Does the company require historical data storage and large-scale integration?

Part 2 : Compare Architectures

1. Create a **comparison table** for Data Warehouse, Data Lake, and Data Lakehouse. Address:
 - Data Types Supported (structured, semi-structured, unstructured)
 - Performance & Scalability
 - Flexibility (handling new data types and evolving needs)
 - Security & Governance
 - Integration with BI and ML Tools

Part 3: Recommendations

1. Based on your analysis, recommend a data architecture for each company:
 - Which architecture suits each company's needs best, and why?
 - What are the trade-offs or limitations of this choice?

Deliverables:

1. A **comparison table** summarizing the architectures.
2. A **recommendation report** explaining and justifying your choices.